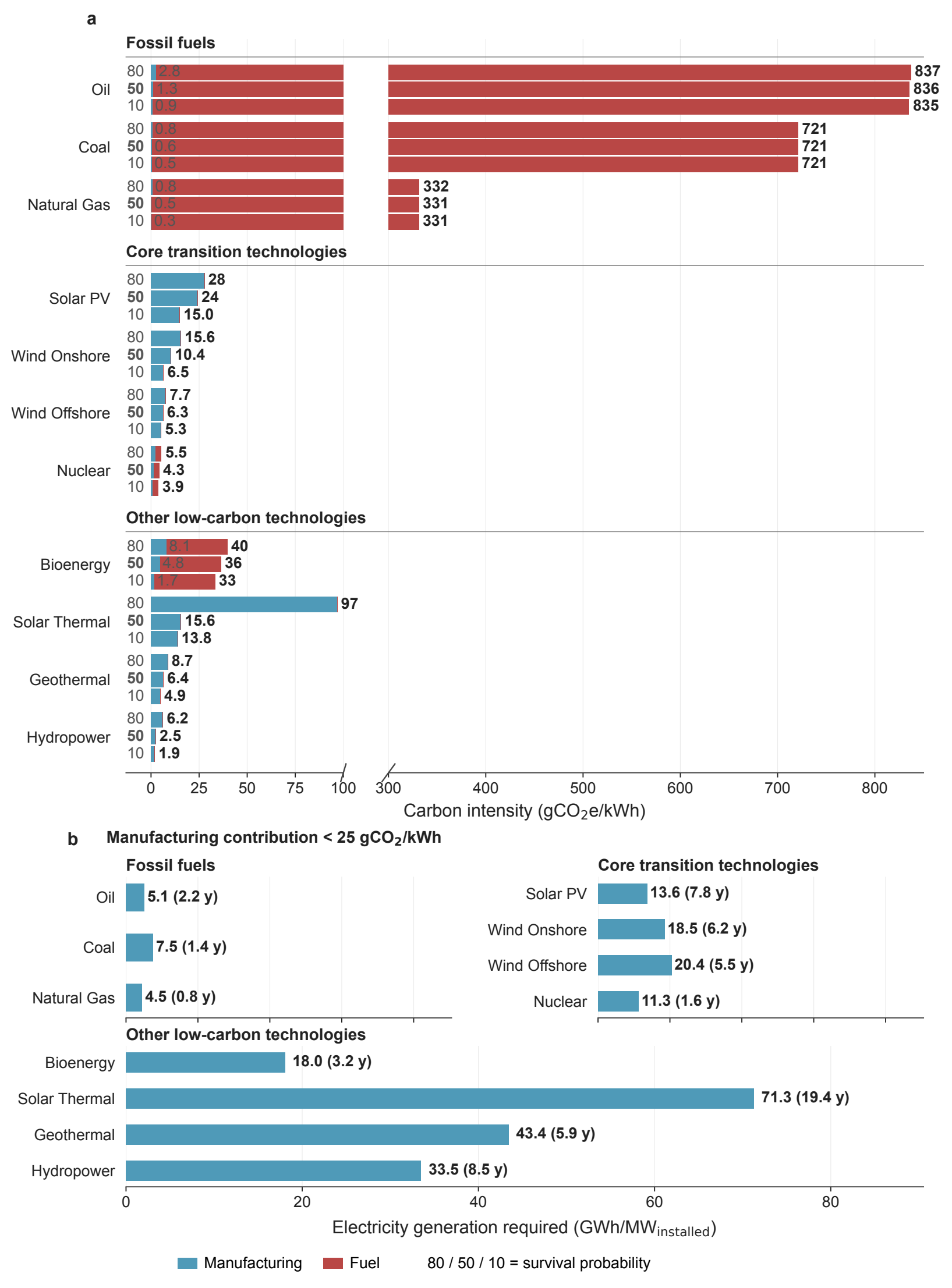

a
Fossil fuels
Oil
80 2.8 837
50 1.3 836
10 0.9 835
Coal
80 0.8 721
50 0.6 721
10 0.5 721
Natural Gas
80 0.8 332
50 0.5 331
10 0.3 331
Core transition technologies
Solar PV
80 28
50 24
10 15.0
Wind Onshore
80 15.6
50 10.4
10 6.5
Wind Offshore
80 7.7
50 6.3
10 5.3
Nuclear
80 5.5
50 4.3
10 3.9
Other low-carbon technologies
Bioenergy
80 8.1 40
50 4.8 36
10 1.7 33
Solar Thermal
80 97
50 15.6
10 13.8
Geothermal
80 8.7
50 6.4
10 4.9
Hydropower
80 6.2
50 2.5
10 1.9
0 25 50 75 100 300 400 500 600 700 800
Carbon intensity (gCO2e/kWh)
b
Manufacturing contribution < 25 gCO2/kWh
Fossil fuels
Oil 5.1 (2.2 y)
Coal 7.5 (1.4 y)
Natural Gas 4.5 (0.8 y)
Core transition technologies
Solar PV 13.6 (7.8 y)
Wind Onshore 18.5 (6.2 y)
Wind Offshore 20.4 (5.5 y)
Nuclear 11.3 (1.6 y)
Other low-carbon technologies
Bioenergy 18.0 (3.2 y)
Solar Thermal 71.3 (19.4 y)
Geothermal 43.4 (5.9 y)
Hydropower 33.5 (8.5 y)
0 20 40 60 80
Electricity generation required (GWh/MWinstalled)
Manufacturing
Fuel
80 / 50 / 10 = survival probability